Chapter 5: Network Layer

Introduction

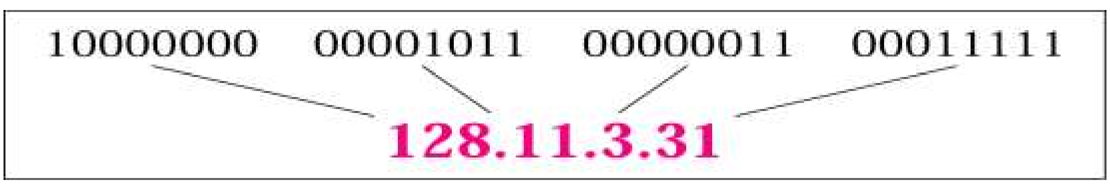
- Layer 3 or Network Layer is backbone of the OSI model
- It finds the best path for data transfer between nodes
- It manages device addressing

A routed protocol is a Layer 3 protocol that applies logical addresses to devices and routes data between networks (such as IP)

A routing protocol dynamically builds the network, topology, and next hop information in routing tables (such as RIP, EIGRP, etc.)

IPV4 address

- 32 bit address
- Total unique address equals to 2^32
    - Around 4.2 billion address
- Represented in dotted Decimal Format
- IANA has authority for IP address management



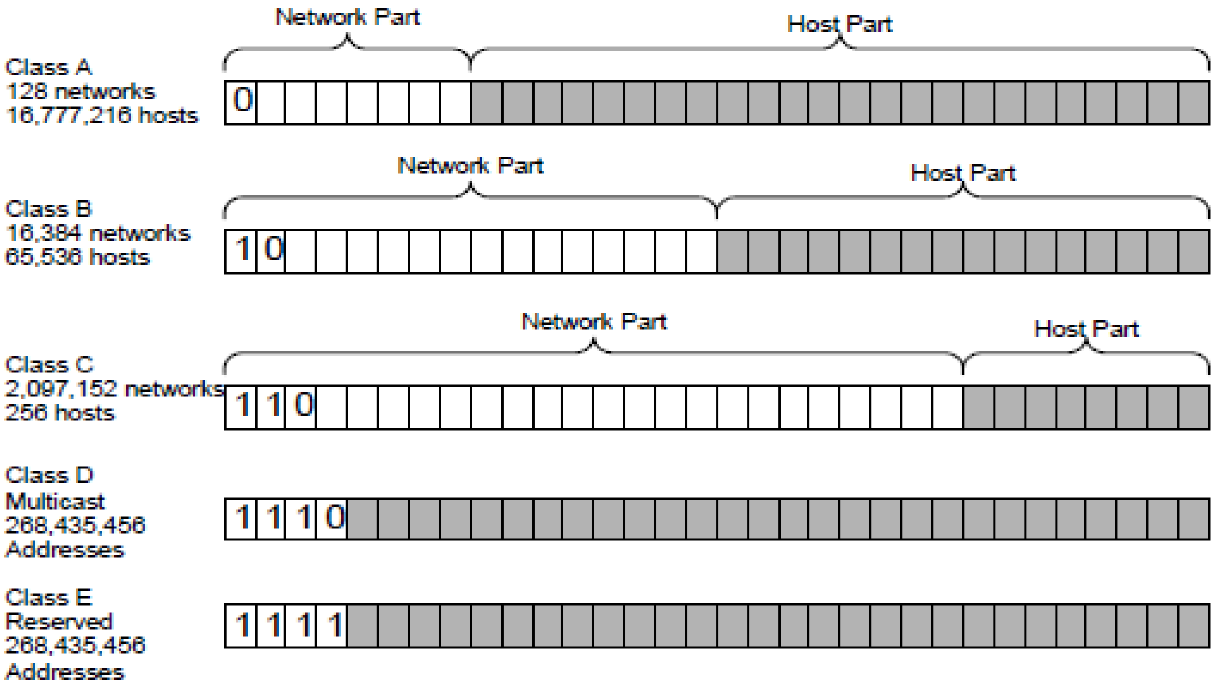
Example of IPV4 address

RIR-Regional Internet Registry manages allocation and registration of internet

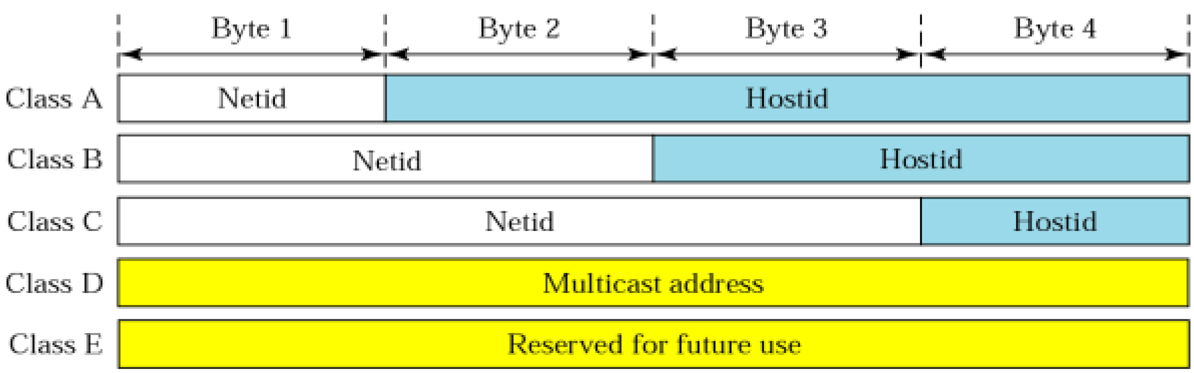
5 RIR are:

- APNIC(Asia Pacific Network Information Centre)
- ARIN (American Registry for Internet Numbers)
- LACNIC (Latin America and Caribbean Network Information Centre)
- AFRINIC (Africa Network Information Centre)
- RIPE NCC (Reseaux IP Europeans Network Coordination Centre)

Classful Addressing



Network Id and Host Id in Classful addressing



Default subnet mask for Classful address

Class	*In Binary*	*In Dotted-Decimal*	*Using Slash*
A	11111111 00000000 00000000 00000000	255.0.0.0	/8
B	11111111 11111111 00000000 00000000	255.255.0.0	/16
C	11111111 111111111 11111111 00000000	255.255.255.0	/24

Logical Address

  - IP address at the Network Layer
  - Used to Communicate with the different subnets

- Netid: Identify network
- Hostid: Identify End devices
- Mask: Used to find netid and hostid
- CIDR: Classless interdomain routing
  - Used in classless addressing
  - Defined by slash notation /n
  - Example: /8, /16, /24

Subnetting and Supernetting

- Subnetting
  - Method used to divide the addresses into several contiguous groups (network)
- Supernetting:
  - Several Network are combined to create a SuperNetwork
  - Mainly used to combine several class C blocks to create a large range of address
  - Supernetting decrease the number of 1s in the mask

Classless Addressing

- To overcome address depletion, classless concept is used
- For classless addressing
  - The address in a block must be contiguous
  - The number of address in a block must be power of 2

Private IP Address

- Range of IP address, which are not routable to internet commonly used for home, office, and enterprise local area networks (LANs)
- If such a private network needs to connect to the Internet, it must use either a network address translator (NAT) gateway, or a proxy server.

Class A	10.0.0.0 – 10.255.255.255
Class B	172.16.0.0 – 172.31.255.255
Class C	192.168.0.0 – 192.168.255.255

Introduction to IPV6

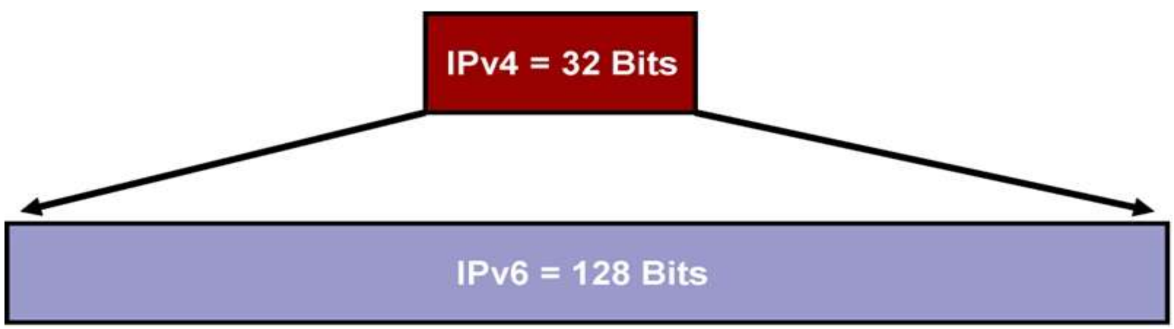
- Major points that played a key role in the birth of IPv6 (drawback of IPV4):
  - address space allowed by IPv4 is saturating.
  - IPv4 on its own does not provide any security features.
    - Data has to be encrypted with some other security application before being sent on the Internet.
  - IPv4 enabled clients can be configured manually or they need some address configuration mechanism.
  - It does not have a mechanism to configure a device to have globally unique IP address.

Features

- Larger Address Space (128 Bit Address Space).
- Supports Resource Allocation via Flow Control Field.
- Supports More Security
- Better Header Format (Base Header and Extension Header)

IPv6 Address

- IPv6: 128 bits or 16 bytes
  - $3.4 * 10^{38}$ possible addressable nodes
  - 340,282,366,920,938,463,374,607,432,768,211,456



128-bit IPv6 Address

Leading Zeros can be removed

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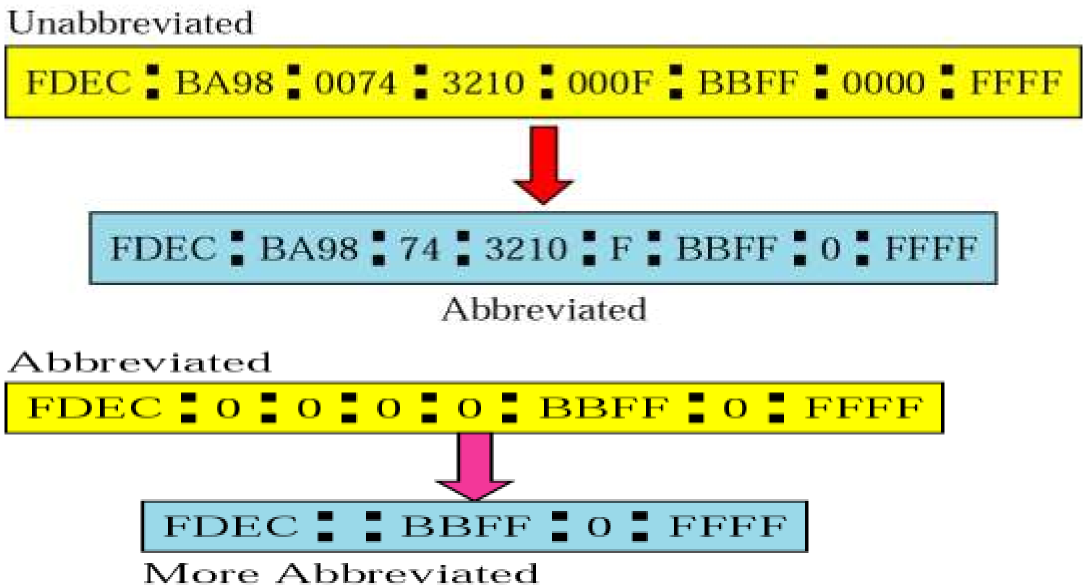


Transition from IPv4 to IPv6

- Tunneling
- Dual Stack
- Header Translation

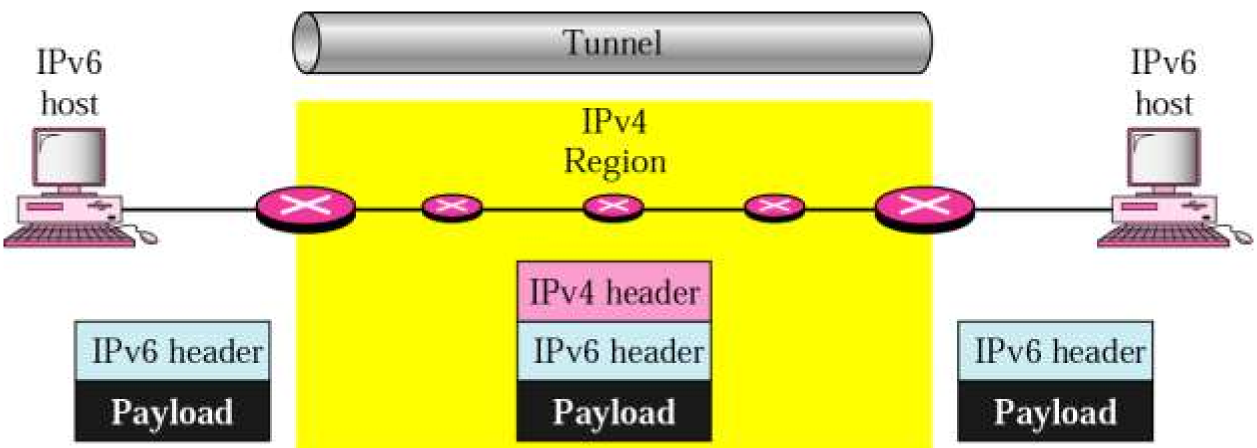


Figure: Tunnelling

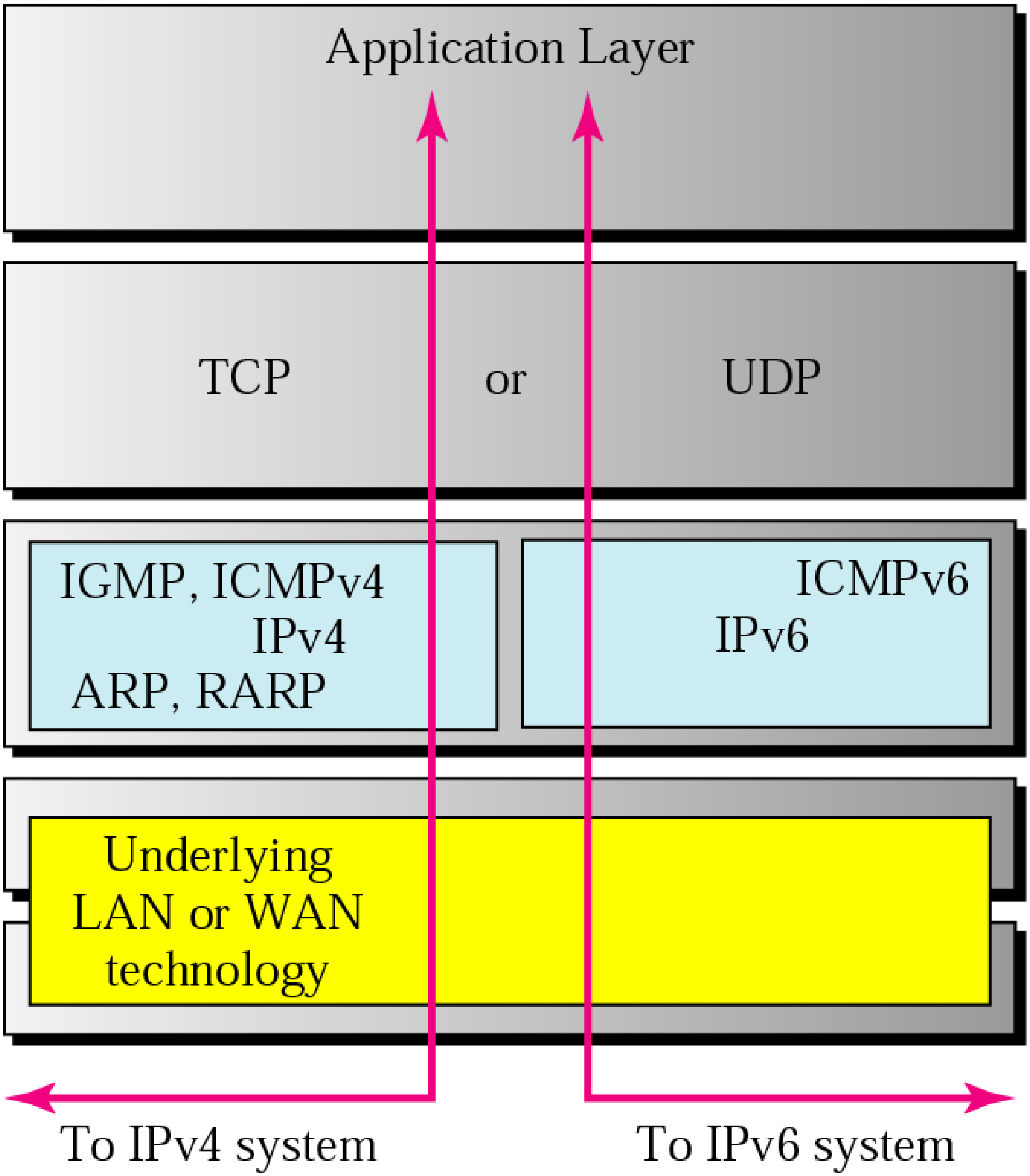


Figure: Dual Stack

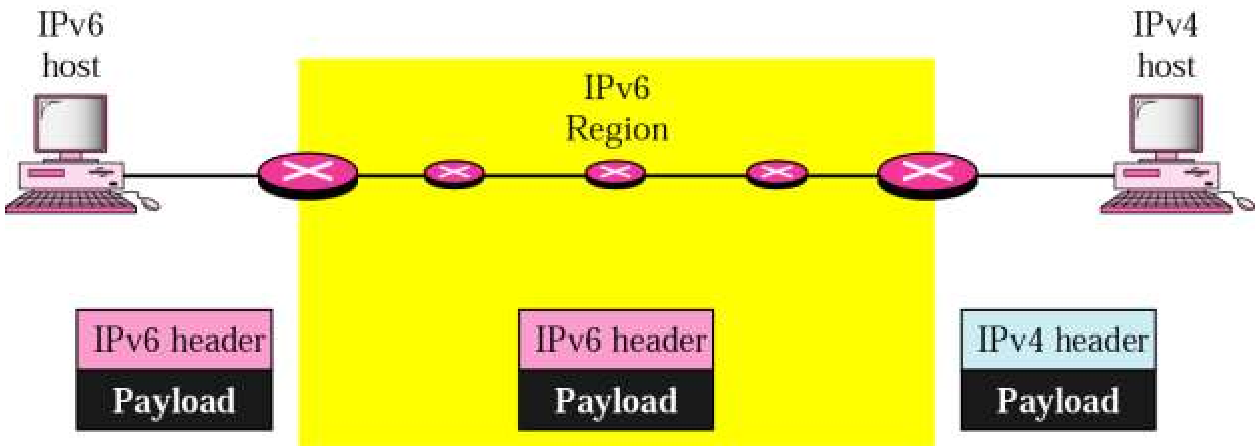


Figure: Header Translation

IPv4 & IPv6 Header Comparison

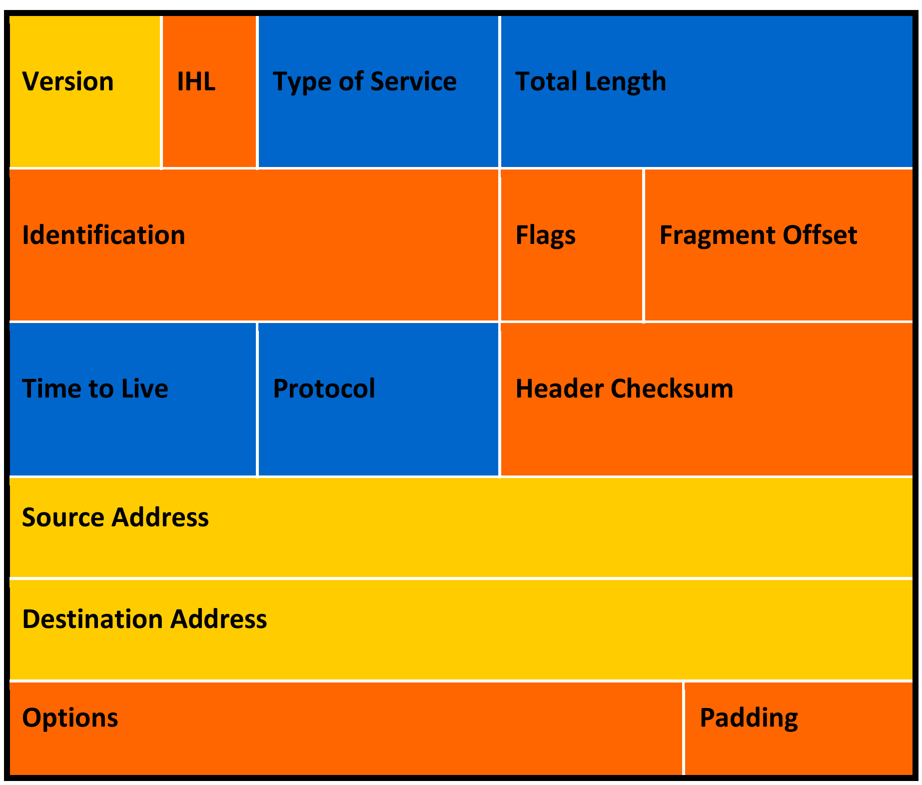


Figure: IPV4 Header

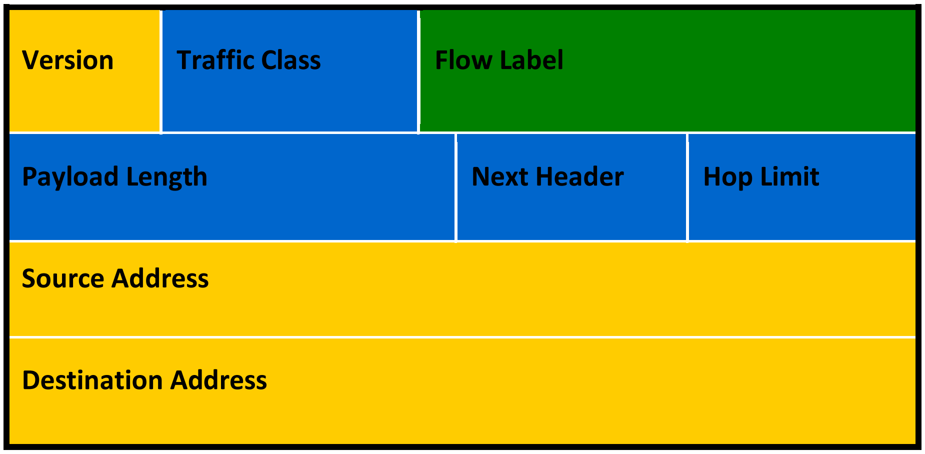


Figure: IPV6 Header

Field's name kept from IPV4 to IPV6

Field's name not in IPV6

Name and position changed in IPV6

New field in IPV6

Comparison of IPV4 and IPV6 Addressing

IPV4	**IPV6**
IPV4 addresses are 32 bit	IPV6 addresses are 128 bit
IPV4 are represented in dotted decimal format	IPV6 addresses are represented in hexadecimal format separated by colon(:)
IPSec support is only optional	Inbuilt IPSec support
Fragmentation is done by senders and forwarding routers	Fragmentation is done by sender
No packet flow identification	Packet Flow identification is available using Flow Label field
Checksum field is available in IPV4 header	No checksum field in IPV6 header
IPV4 configuration is done manually or using DHCP	Auto-configuration addresses is available

Options field is available	No options field but IPV6 extension header are available
Limited address space	Large addressing space
Broadcast address are available	No broadcast, its function is superseded by multicast address

Routing Algorithm Overview

Classful Routing

- Classful routing protocols do not send the subnet mask along with their updates
- Examples of Classful routing protocol is RIP version 1

Classless Routing Protocols

- Classless Routing Protocols send the Subnet mask along with their updates
- Benefits for using Classless routing protocols
  - ➢ We can save lot of IP address using VLSM techniques in classless routing protocols
  - ➢ Examples: OSPF, EIGRP, RIPV2

Adaptive Routing Protocols

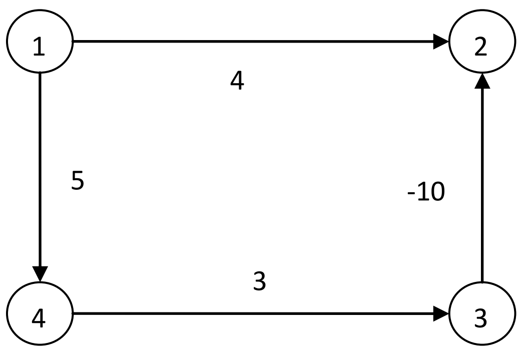
- Adaptive routing protocols can easily adapt to network changes.
- Routing protocols are used by routers to inform one another of the IP networks accessible to them.
- Adaptive Routing protocols are also called Dynamic routing Protocols
- Examples of Adaptive routing Protocols are: OSPF, EIGRP, RIP, BGP, IGRP

Non-Adaptive Routing Protocols

- Non Adaptive Routing Protocols doesn't respond to change in topology
- Non-Adaptive Routing Protocols are also called as Static routing Protocols
- Non-Adaptive Routing protocols are more secure than Adaptive Routing protocols as they don't advertise the routing update unnecessarily
- Network Administrator is responsible for configuring the static routing in the router
- Static routes are fixed and do not change if the network is changed or reconfigured

Distance Vector Routing

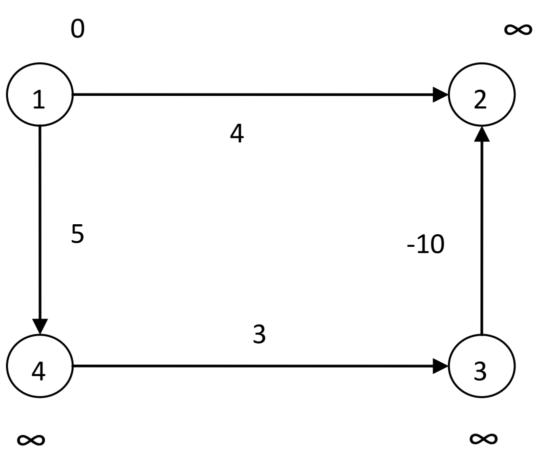
It is a dynamic routing algorithm. Distance vector routing algorithms operate by having each router maintain a table (i.e, a vector) giving the best known distance to each destination and which line to use to get there. These tables are updated by exchanging information with the neighbors. It is also called Bellman-Ford routing algorithm.

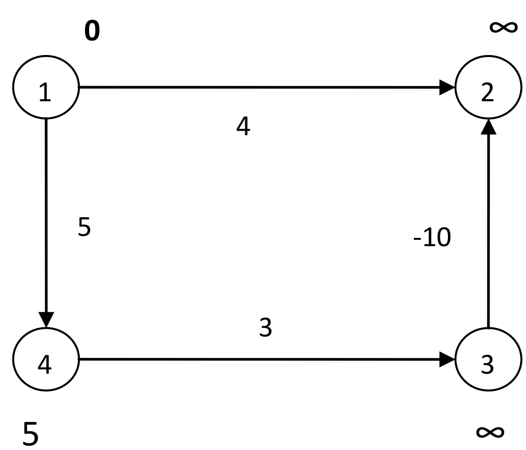


Let us consider source vertex 1, then we need to find the shortest path to all other vertices or nodes. For n vertices, n-1 iteration gives the shortest path, but we may get result in less than n-1 as well. Since there are 4 vertices we need 3 iteration to find the shortest path.

Edges-> (3,2),(4,3),(1,4),(1,2)

Iteration 1





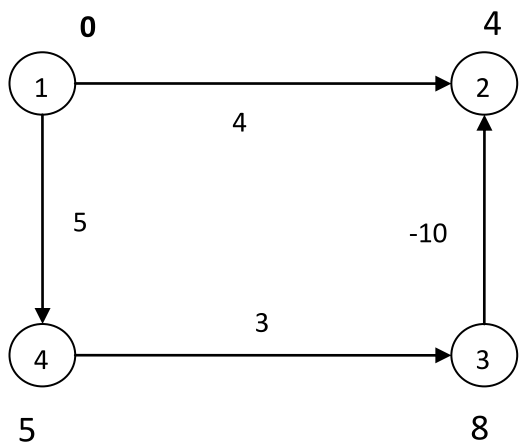
Initially the initial vertex is marked as 0 and other as infinity, then we find the cost to every vertex according to edges as: (3, 2) -> ∞ - 10 =∞

(4, 3) -> ∞+3=∞

(1, 4) ->0+5=5 (less than infinity so take this value)

(1,2) ->0+4=4 (less than infinity so take this value) [take value if less than given value]

Iteration 2



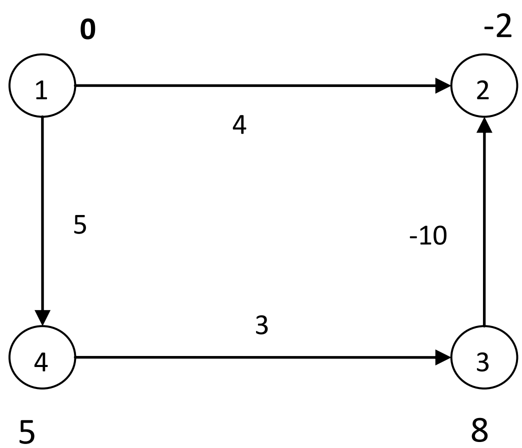
(3,2)-> ∞ - 10 =∞

(4,3)-> 5+3=8

(1,4)->0+5=5

(1,2)->0+4=4

Iteration 3



(3,2)-> 8 - 10 =-2

(4,3)-> 5+3=8

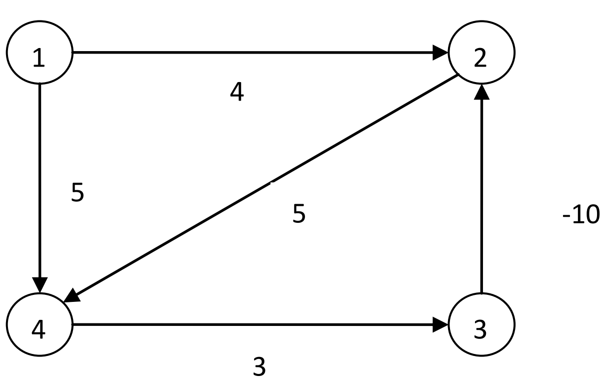
(1,4)->0+5=5

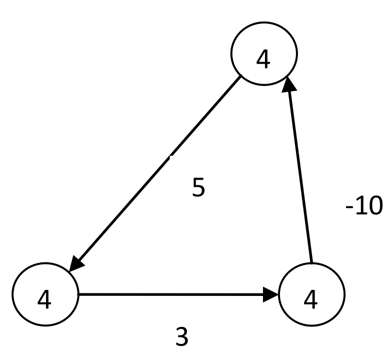
(1,2)->0+4=-2

Change value only if you get lesser value in the iterations.

So shortest path from 1 to other vertices are: 1=0, 2=-2, 3=8 and 4=5

Drawbacks





If there is negative weight with cycle, then we don't get result in n-1 iteration it could go on and we can see in above example, but it works with positive result.

Link State Routing

- Distance vector routing algorithm such as rip has limitations that they don't work for longer network
- Convergence and scalability problem for topology changes
- Link state routing is used to overcome those limitations
  - Link state routing has full knowledge of network.
  - Each node maintains the full graph by collecting the updates from all other nodes
  - Each node then independently calculates the next best logical *path* from it to every possible destination in the network
  - Router's receive topology information from their neighbor router via link state advertisements(LSA)
  - Use Dijkstra's shortest path first algorithm to find out optimal paths
  - Link state protocols don't have to constantly resend their entire LSAs instead they can send small hello LSAs to let their neighbor routers know they are still alive
- The idea behind link state routing is fairly simple and can be stated as five parts. Each router must do the following:
  - Discover its neighbors, learn their network address.
  - Measure the delay or cost to each of its neighbors.
  - Construct a packet telling all it has just learned.
  - Send this packet to all other routers.
  - Compute the shortest path to every other router.

Interior Routing Protocols

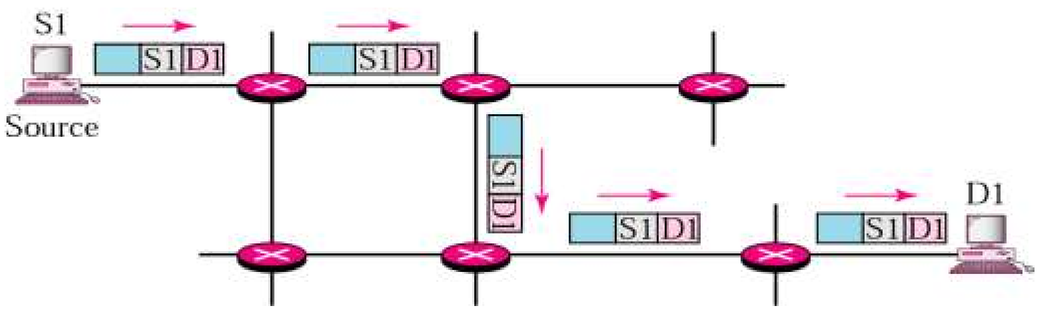
- Used for Routing Inside an Autonomous System (AS).
- Used within the Organization
- AS => Network under Common Administration.
- Router having Same AS, share their routing tables
- Examples => RIP, EIGRP and OSPF, IGRP

Exterior Routing Protocols

- Used for Routing between Autonomous System (AS)
- Border Gateway Routing Protocols (BGP)
- Used between the organization (ISPs to ISPs)

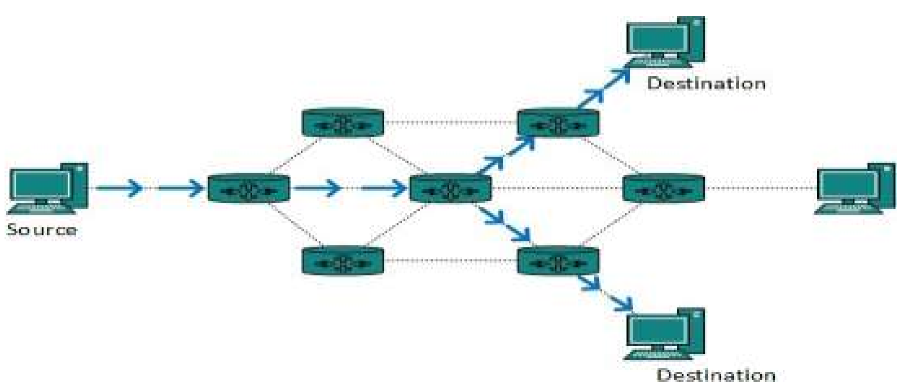
Unicast Routing

- In unicasting there is a single sender (source) and a single receiver (destination)
- In unicast routing, the router forwards the received packet through only one of its interfaces
- Examples of Unicast Routing are: OSPF, RIP and BGP



MultiCast Routing

- In multicasting there is at least one sender and several receivers (group of receivers called multicast group)
- In multicast routing, the router may forward the received packet through several of its interfaces.
- M-cast group address "delivered" to all receivers in the group
- Internet uses Class D for m-cast
- M-cast address distribution, etc. managed by IGMP Protocol



RIP (Routing Information Protocol)

RIP is a routing protocol for exchanging routing table information between routers. Routing updates must be passed between routers so that they can make the proper choice on how to route a packet. It is oldest Distance Vector Routing Protocol. It update routing table every 30 sec. It uses Hop count as Metric to find best path to the destination. RIP has a maximum hop count of 15. A route with a hop count greater than 15 is considered unreachable. It has two version:RIP Version 1 and RIP Version 2. RIP version 1 is the classful routing protocol. RIP Version 2 is classless routing protocol. In fact, most DSL/cable modem routers such as the ones from Linksys come bundled with RIP.

OSPF (Open Shortest Path First)

It is a Classless routing protocol. It uses a link state routing algorithm and falls into the group of interior routing protocols. OSPF was developed as a replacement for the distance vector routing protocol RIP. It uses cost to find the best route to find the destination

Every intra-domain must have a core area

- It is referred to as a *backbone area*
- This is identified with Area ID 0
- Areas are identified through a 32-bit area field
- Thus Area ID 0 is the same as 0.0.0.0

Areas (other than the backbone) are sequentially numbered as Area 1 (i.e., 0.0.0.1), Area 2, and so on

BGP (Border Gateway Routing Protocols)

It is designed to exchange routing and reach-ability information between autonomous systems (AS) on the Internet. It is of two types: Internal BGP and External BGP. Internal BGP has the Administrative Distance of 200 while external BGP has the Administrative Distance of 20. The protocol is often classified as a path vector protocol, but is sometimes also classed as a distance-vector routing protocol.

PATH VECTOR PROTOCOL

It is a routing protocol which maintains the path information that gets updated dynamically.

Updates which have looped through the network and returned to the same node are easily detected and discarded. Peers exchange BGP messages using TCP. BGP defines 4 types of messages:

- OPEN: opens a TCP connection to peer and authenticates sender
- UPDATE: advertises new path (or withdraws old)
- KEEPALIVE: keeps connection alive in absence of UPDATES; also serves as ACK to an OPEN request
- NOTIFICATION: reports errors in previous message; also used to close a connection